

Yaqin Chen

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RESEARCH INTERESTS

Primary Areas: Machine Learning, AI for Science, Sequential Decision-Making

Specific Topics: Deep Generative Models, Reinforcement Learning, Optimization, Statistical Machine Learning

EDUCATION

School of Mathematics (Zhuhai), Sun Yat-sen University

Sep 2022 – Jun 2026 (expected)

- B.S. in Information and Computing Science, GPA: 4.0/4.0 (1/60)

- Core curriculum: Mathematical Statistics, Real Analysis, Numerical Analysis, Stochastic Processes, Optimization

Department of Statistics, University of Wisconsin-Madison

Sep 2025 – May 2026 (expected)

- Visiting International Student Program (Thematic Statistics Track)

- Core curriculum: Statistical Inference, Statistical Methods, Machine Learning and Statistical Pattern

RESEARCH EXPERIENCE

Decision, Risk and Operations Division, Columbia Business School *Research Intern*

Mar 2025 – Present

Advisor: [Hongseok Namkoong](#)

- Researched scalable benchmarks for evaluating large language models in interactive, multi-turn environments.
- Researched the in-context experiential learning capabilities of frontier language models across repeated interactions.
- Researched the simulation of user behaviors to identify and analyze emergent failure modes in LLM agents.

iSURE Program, University of Notre Dame *Research Intern*

July 2025 – Oct 2025

Advisor: [Jianxun Wang](#) ([Sibley School of Mechanical Engineering, Cornell University](#))

- Researched the theoretical and mathematical unification of diffusion models and flow-matching generative frameworks.
- Adapted and benchmarked posterior sampling for flow-based conditional generation on scientific datasets (e.g., turbulence).
- Investigated training-free guidance techniques to accelerate sampling speed while ensuring generation accuracy.

Human Cyber Physical Intelligence Laboratory, Sun Yat-sen University *Research Intern*

Feb 2024 – Feb 2025

Advisor: [Pengxu Wei](#), [Liang Lin](#) ([Fellow of IEEE/IAPR](#))

- Researched the impact of latent alignment and structural inductive biases in Variational Autoencoders (VAEs).
- Researched a statistical framework for efficient representation learning with Minimum Variance Unbiased Estimator.
- Researched novel autoencoder architectures that enforce statistical efficiency to improve representation learning quality.

PUBLICATIONS

Benchmarking In-context Experiential Learning Through Repeated Product Recommendations[\[pdf\]](#)

Gilbert Yang*, **Yaqin Chen***, Thomson Yen, Hongseok Namkoong

Under review at the International Conference on Learning Representations (ICLR), 2026

Training-Free Guided Generation techniques for Conditional Flow Matching on Scientific Datasets[\[pdf\]](#)

Meet Hemant Parikh, **Yaqin Chen**, Jianxun Wang

Under review at xxxxx

MVU-AE: Minimum Variance Unbiased Autoencoders[\[pdf\]](#)

Yaqin Chen, Junrong Lian, Qing Zhao, Xiangyang Li, Liang Lin, Pengxu Wei

SELECTED PROJECTS

Development of a Martial Arts Teaching and Scoring System Based on YOLOv8

[Demo Pages](#)

2024 Chinese Academy of Sciences Innovation and Practice Training Program For Undergraduate

Project Leader

- Developed a real-time martial arts coaching system using YOLOv8 for pose estimation and movement analysis.
- Engineered an automated scoring engine based on video analysis to provide objective performance feedback.
- Integrated a Large Language Model (LLM) to generate personalized, multi-modal (text and voice) coaching.

AWARDS

Chinese National Scholarship (Top 1%)	2024
First-Class Excellent Student of SYSU (Top 5%, 1/151)	2024
First-Class Excellent Student of SYSU (Top 5%, 1/151)	2023
MCM Modeling Contest Meritorious Award (7% of 11296)	2023
First Prize 37th Chinese Physics Olympiad of Jiangxi Province (Top 5%)	2020

PROFESSIONAL SKILLS

Programming Languages: Python, C++, R, MATLAB

Languages: English(fluent), Chinese(native)

STANDARDIZED TESTS

GRE: Verbal 129 (80%) +Quantitative 170 (92%) + Analytical Writing 3

TOEFL: Total 108 (Reading 28+Listening 27+Speaking 26+Writing 27)